Stem society

MATH AND SCI: ENIGMA

The Message Summit ; Rebirth Of the Phoenix

INRTODUCTION

STEM ENIGMA CHALLENGES STUDENTS FROM GRADES 8–11 TO APPLY THEIR MATHEMATICAL REASONING, SCIENTIFIC KNOWLEDGE AND LOGICAL PROBLEM-SOLVING SKILLS. PARTICIPANTS WILL WORK IN MIXED-SUBJECT TEAMS DURING THE INITIAL ROUND, COMBINING MATHEMATICS AND INTEGRATED SCIENCES TO INVESTIGATE AND SOLVE CHALLENGING CASE FILES. QUALIFYING PARTICIPANTS WILL THEN ADVANCE TO AN INDIVIDUAL SCAVENGER HUNT, WHERE THEY WILL PUT THEIR SUBJECT-SPECIFIC KNOWLEDGE AND QUICK THINKING TO THE TEST. THIS STUDY GUIDE OUTLINES THE KEY CONCEPTS AND SKILLS REQUIRED TO SUCCEED IN BOTH ROUNDS.

STUDY MATERIAL

MATHEMATICS: CORE TOPICS

1. NUMBER SYSTEMS AND ARITHMETIC: NUMBER TYPES, POSITIVE AND NEGATIVE NUMBERS, FRACTIONS, DECIMALS, PERCENTAGES, BODMAS, FACTORS, MULTIPLES, HCF, LCM, PRIME FACTORISATION, POWERS, ROOTS, STANDARD FORM, RATIOS, PROPORTION, ROUNDING AND SIGNIFICANT FIGURES.
2. ALGEBRA: SIMPLIFICATION, EXPANSION, FACTORISATION, ALGEBRAIC IDENTITIES, SUBSTITUTION, LINEAR AND SIMULTANEOUS EQUATIONS, QUADRATIC EQUATIONS, SEQUENCES, NTH-TERM RULES AND INEQUALITIES.
3. GEOMETRY: TYPES OF ANGLES, ANGLE PROPERTIES, PARALLEL LINES, CORRESPONDING ANGLES AND VERTICALLY OPPOSITE ANGLES.
4. MENSURATION: PERIMETER, AREA, CIRCUMFERENCE, SURFACE AREA AND VOLUME OF 2D AND 3D SHAPES.
5. TRIGONOMETRY: PYTHAGORAS' THEOREM, SINE, COSINE AND TANGENT RATIOS, AND FINDING UNKNOWN SIDES AND ANGLES IN RIGHT-ANGLED TRIANGLES.
6. GRAPHS AND FUNCTIONS: CARTESIAN COORDINATES, PLOTTING AND INTERPRETING GRAPHS, GRADIENTS, EQUATIONS OF STRAIGHT LINES AND QUADRATIC GRAPHS.
7. STATISTICS AND PROBABILITY: MEAN, MEDIAN, MODE, RANGE, BAR CHARTS, PIE CHARTS AND HISTOGRAMS.
8. MATHEMATICAL REASONING AND PROBLEM-SOLVING: NUMBER AND SHAPE PATTERNS, LOGICAL REASONING, MISSING NUMBERS, MATHEMATICAL RIDDLES, MENTAL MATHEMATICS AND APPLICATION OF MATHEMATICAL CONCEPTS TO UNFAMILIAR PROBLEMS.

STUDY MATERIAL

LOGICAL STEM UNDERWORLD: CORE SCIENCE TOPICS

A. BIOLOGY

NOT COMPULSORY FOR NON-BIOLOGY STUDENTS.

CELLS AND ORGANISATION: CELL STRUCTURE, ORGANELLES, SPECIALISED CELLS AND LEVELS OF ORGANISATION.

HUMAN BODY SYSTEMS: DIGESTIVE, RESPIRATORY AND CIRCULATORY SYSTEMS.

PHOTOSYNTHESIS: PROCESS, REQUIREMENTS AND FACTORS AFFECTING ITS RATE.

RESPIRATION AND TRANSPORT: AEROBIC AND ANAEROBIC RESPIRATION, GAS EXCHANGE AND PLANT TRANSPORT.

REPRODUCTION: SEXUAL AND ASEXUAL REPRODUCTION, POLLINATION AND PLANT REPRODUCTION.

B. CHEMISTRY

STATES OF MATTER: PROPERTIES, CHANGES OF STATE, PARTICLE THEORY AND DIFFUSION.

ATOMIC STRUCTURE: SUBATOMIC PARTICLES, ATOMIC STRUCTURE, ELEMENTS AND PERIODIC TABLE.

CHEMICAL BONDING: IONIC AND COVALENT BONDING, FORMATION OF IONS.

CHEMICAL REACTIONS: EQUATIONS, BALANCING, REACTION TYPES AND ENERGY CHANGES.

ACIDS, BASES AND SALTS: PH, INDICATORS, NEUTRALISATION AND SALT FORMATION.

METALS AND REACTIVITY: PROPERTIES, REACTIVITY SERIES AND DISPLACEMENT REACTIONS.

SEPARATION TECHNIQUES: FILTRATION, EVAPORATION, CRYSTALLISATION, DISTILLATION AND CHROMATOGRAPHY.

C. PHYSICS

MOTION AND FORCES: SPEED, VELOCITY, ACCELERATION, GRAPHS, NEWTON'S LAWS, MASS, WEIGHT AND MOMENTUM.

ENERGY: ENERGY FORMS, TRANSFERS, CONSERVATION, RESOURCES AND EFFICIENCY.

WORK, POWER AND PRESSURE: WORK DONE, POWER, PRESSURE AND APPLICATIONS.

WAVES, LIGHT AND SOUND: WAVE PROPERTIES, FREQUENCY, WAVELENGTH, REFLECTION AND REFRACTION.

HEAT AND TEMPERATURE: THERMAL EXPANSION, HEAT TRANSFER, CHANGES OF STATE AND SPECIFIC HEAT CAPACITY.

CATEGORY OVERVIEW

DIGICON :

THE CATEGORY CONSISTS OF TWO MAIN ROUNDS: SQUAD BATTLE (TEAM ROUND) AND SOLO SHOWDOWN (INDIVIDUAL ROUND). PARTICIPANTS WILL FIRST COMPETE IN TEAMS THROUGH FOUR INTERACTIVE STATIONS, WITH THE TOP-PERFORMING TEAMS ADVANCING TO THE SECOND ROUND. THE FINAL ROUND WILL BE AN INDIVIDUAL WRITTEN COMPETITION, WHERE PARTICIPANTS WILL EARN CODE POINTS BASED ON THEIR PERFORMANCE. THE COMPETITION CARRIES A TOTAL OF 100 CODE POINTS, AND THE TWO PARTICIPANTS WITH THE HIGHEST OVERALL SCORES WILL BE DECLARED THE WINNERS

ENIGMA :

.THE CATEGORY CONSISTS OF TWO MAIN ROUNDS: INVESTIGATING CASE FILES (TEAM ROUND) AND SCAVENGER HUNT STYLE QUIZ (INDIVIDUAL ROUND). PARTICIPANTS WILL INITIALLY REGISTER INDIVIDUALLY FOR MATHEMATICS OR INTEGRATED SCIENCES AND WILL THEN BE GROUPED INTO MIXED-SUBJECT TEAMS TO SOLVE A CASE FILE USING MATHEMATICAL AND SCIENTIFIC KNOWLEDGE. THE TOP-PERFORMING TEAMS WILL ADVANCE TO THE FINAL ROUND, WHERE EACH QUALIFYING PARTICIPANT WILL COMPETE INDIVIDUALLY IN A SUBJECT-SPECIFIC SCAVENGER HUNT BY SOLVING QUESTIONS WITHIN A GIVEN TIME LIMIT. MATHEMATICS AND INTEGRATED SCIENCES WILL HAVE SEPARATE COMPETITIONS, WITH A WINNER AND RUNNER-UP DECLARED FOR EACH CATEGORY.